AI ACADEMY · PYTHON READINESS BRIDGE · STUDENT LESSON

Loops and trace tables

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

Before you start

Starting with total = 2, what does total = total + 3 store?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Trace variable values before and after each loop iteration. Try the recall before opening notes.

Your learning target

I can trace variable values before and after each **loop iteration**.

I can use the excluded stop value in range correctly.

I can explain **initialization** and test zero one and several **iteration**s.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Branching and boundary tests. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Expressions and operator order. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Trace variable values before and after each **loop iteration**

Use the excluded stop value in range correctly

Explain **initialization** and test zero one and several **iteration**s

Retrieve what you already know

> Starting with total = 2, what does total = total + 3 store?

My starting explanation

This week's end goal

Trace **loop**s and **trace table**s and explain the changed program before running it.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Loops and trace tables case. Separate observations, given facts and assumptions before choosing an explanation.

Imagine adding numbered counters: first 1, then 2, then 3. A **loop** can repeat the same addition with a different number each time. This week you will predict each change before running a short Python script.

Tracing repeated calculations prepares you to inspect the data **process**ing used in larger AI applications.

In for turn in range(1, 4), Python supplies 1, 2 and 3 in order. The start is included and the stop is excluded. The indented body runs once for each supplied value. The colon and four-space indentation mark that body.

This lesson uses integer range values with the default positive step of 1.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Loops and trace tables. Connect each labelled input, deciding step and result.

```
total = 0
for turn in range(1, 4):
    total = total + turn
print(total)
```

State after each addition

TURN	TOTAL BEFORE	TOTAL AFTER
1	0	1
2	1	3
3	3	6

Initialize total to 0 before the loop. The unindented print runs once afterward.

The **output** is 6. Before the first **iteration**, total is 0. Each assignment reads the previous total and adds the current turn. After turn 3, the range is exhausted, so execution continues after the **loop**.

The table records state; it does not create extra instructions.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Loops and trace tables. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Replace range(1,5) with range(1,1).

Predicted result

Prediction: No values are visited and the final **output** is 0.

Why the result follows

Reason: The **accumulator** starts at zero and an empty **loop** never changes it.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Loops and trace tables to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Loop	A block of instructions that can run repeatedly
Iteration	One execution of a loop body
Accumulator	A variable that keeps a running result
Initialization	Giving a variable its starting value
Trace table	A record of variable values as instructions execute

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Loops and trace tables, predict a step, then compare with its explanation.

Range expression	Values supplied	Number of iterations
range(1, 4)	1, 2, 3	3
range(1, 2)	1	1
range(1, 1)	No values	0
range(4)	0, 1, 2, 3	4

Worked explanation · Loops and trace tables

With one argument, range starts at zero. range(4) supplies four values but none is 4. If the worked script uses range(1, 1), the body never runs and the initialized total remains 0.

Do not rely on a variable first assigned inside a **loop**: a zero-**iteration** run might never assign it.

Replace range(1, 4) with range(1, 3) in the worked script. Predict the supplied values and final total. What stop value is needed to include 3?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Loops and trace tables

Accumulate a short range. A running total adds the integers visited by range(1,5).

The facts we will use

Supplied facts

Python 3: total=0; for value in range(1,5): total+=value; print(total). The print is after the **loop**.

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Running total after each visited value

	Current value	Running total
Visit 1	1	1
Visit 2	2	3
Visit 3	3	6
Visit 4	4	10

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / PYTHON-BRIDGE/6

Loops and trace tables

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Visit the range

Values are 1,2,3,4; the stop value 5 is excluded.





2. Trace running total

Totals after each addition are 1,3,6,10.





3. Print after the loop

The final displayed value is 10.

CHANGE ONE THING

Replace range(1,5) with range(1,1).

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Visit the range	Values are 1,2,3,4; the stop value 5 is excluded.
2. Trace running total	Totals after each addition are 1,3,6,10.
3. Print after the loop	The final displayed value is 10.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **loop** repeats an operation for each visited value. A **trace table** distinguishes the current value from the **accumulator**; the range stop is not included.

What this example does not establish

Keep the stated limit

This bounded **loop** does not train an AI model; other **loop** conditions may require separate termination checks.

Keep the stated limit

Read the labelled picture: List the four visited values and their running totals. Explain why 5 is absent and what prints for range(1,1).

Keep the stated limit

Change one thing: Replace range(1,5) with range(1,1). Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Loops and trace tables claim and explain a repair using the supplied evidence.

```
total = 0
for turn in range(1, 4):
    total = 0
    total = total + turn
print(total)
```

This faulty version prints 3. The extra reset discards earlier additions on every **iteration**. Remove the indented reset while keeping **initialization** before the **loop**.

Then test zero, one and several **iteration**s: the one-**iteration** case alone cannot reveal this reset bug.

Moving print(total) into the correct **loop** produces three lines, 1 then 3 then 6. Moving it out produces only the final 6. Trace both values and **output** events.

A **loop** follows written instructions; repeating them does not make the program learn.

Locate the earliest unsupported step

For the faulty script with range(1, 4), record total after each addition. Explain the wrong final **output** and remove only the faulty reset.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Loops and trace tables with support, then explain the deciding step in your own words.

List the values supplied by range(2, 5) and explain why 5 is absent. How many times does its body run?

Replace range(1, 4) with range(1, 3) in the worked script. Predict the supplied values and final total. What stop value is needed to include 3?

Explain your prediction to a partner before comparing results.

Practise with less help: Trace before running

Use workbook W2 for the connected example "**Loop**s and **trace table**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Loop**s and **trace table**s. Proposed explanation: "The stop value is included". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

List the four visited values and their running totals. Explain why 5 is absent and what prints for range(1,1).

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Loops and trace tables. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

```
total = 0
for turn in range(1, 4):
    total = total + turn
print(total)
```

```
total = 0
for turn in range(1, 4):
    total = 0
    total = total + turn
print(total)
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	

Record	Your evidence
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Loops and trace tables. Explain why each step is justified.

Explain → apply → challenge

> List the values supplied by range(2, 5) and explain why 5 is absent. How many times does its body run?

For the faulty script with range(1, 4), record total after each addition. Explain the wrong final **output** and remove only the faulty reset.

In your counter script, range(-2) performs no **iteration**s. Does returning the initial 5 make negative rounds valid? Explain what the script checks and what it assumes.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Python readiness portfolio. Keep an annotated decision and the evidence supporting it.

Add a labelled **loop**s and **trace table**s **evidence** sheet to Python readiness portfolio. Show how you can trace variable values before and after each **loop iteration**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Write the expected range values and **trace table** before running. Keep predicted and observed **output** separate, then explain the first row where they differ.

For a **loop** that should add a fixed amount repeatedly, zero **iteration**s should preserve the initial amount. One **iteration** checks one update; several check accumulation.

> **A learner checks the reset bug using only range(1, 2) and sees the expected answer 1. Explain why this does not establish correctness, and give a revealing test with its expected correct output.**

In your counter script, range(-2) performs no **iteration**s. Does returning the initial 5 make negative rounds valid? Explain what the script checks and what it assumes.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Loops and trace tables case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

State after each addition

TURN	TOTAL BEFORE	TOTAL AFTER
1	0	1
2	1	3
3	3	6

Initialize total to 0 before the loop. The unindented print runs once afterward.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Count iterations and a stopping point

List every value visited and each running total. Then predict the result for range(2,2).

```
total = 0
for value in range(2, 6):
    total = total + value
print(total)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the **loop**.

Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

Your lesson route

> Use this map to navigate Loops and trace tables, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Accumulate a short range. A running total adds the integers visited by range(1,5).

By the end, I can	How I will show it
Trace variable values before and after each loop iteration	Explain the deciding step, show a trace or test, and state one limit.
Use the excluded stop value in range correctly	Explain the deciding step, show a trace or test, and state one limit.
Explain initialization and test zero one and several iterations	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Loops and trace tables without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Trace variable values before and after each **loop iteration**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Accumulate a short range. A running total adds the integers visited by range(1,5).

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The final displayed value is 10.

1. Trace running total

2. Print after the **loop**

3. Visit the range

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Print after the **loop** The final displayed value is 10. A **loop** repeats an operation for each visited value.

A **trace table** distinguishes the current value from the **accumulator**; the range stop is not included.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace range(1,5) with range(1,1). Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. No values are visited and the final **output** is 0.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: No values are visited and the final **output** is 0. The **accumulator** starts at zero and an empty **loop** never changes it.

Repair a missing building block

Review Branching and boundary tests (python-bridge/5). Goal: Trace the selected branch in an if elif else chain. Short practice: Trace -2, 4 and 5 through the branches.

Explain why the negative check comes first and what label -2 gets if it is removed.. Return to this week and explain what you changed.

Review Expressions and operator order (python-bridge/3). Goal: Evaluate numeric expressions using Python’s grouping rules. Short practice: Trace the operation order in both expressions.

Identify which picture step changes with parentheses and predict both results when the multiplier becomes four.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

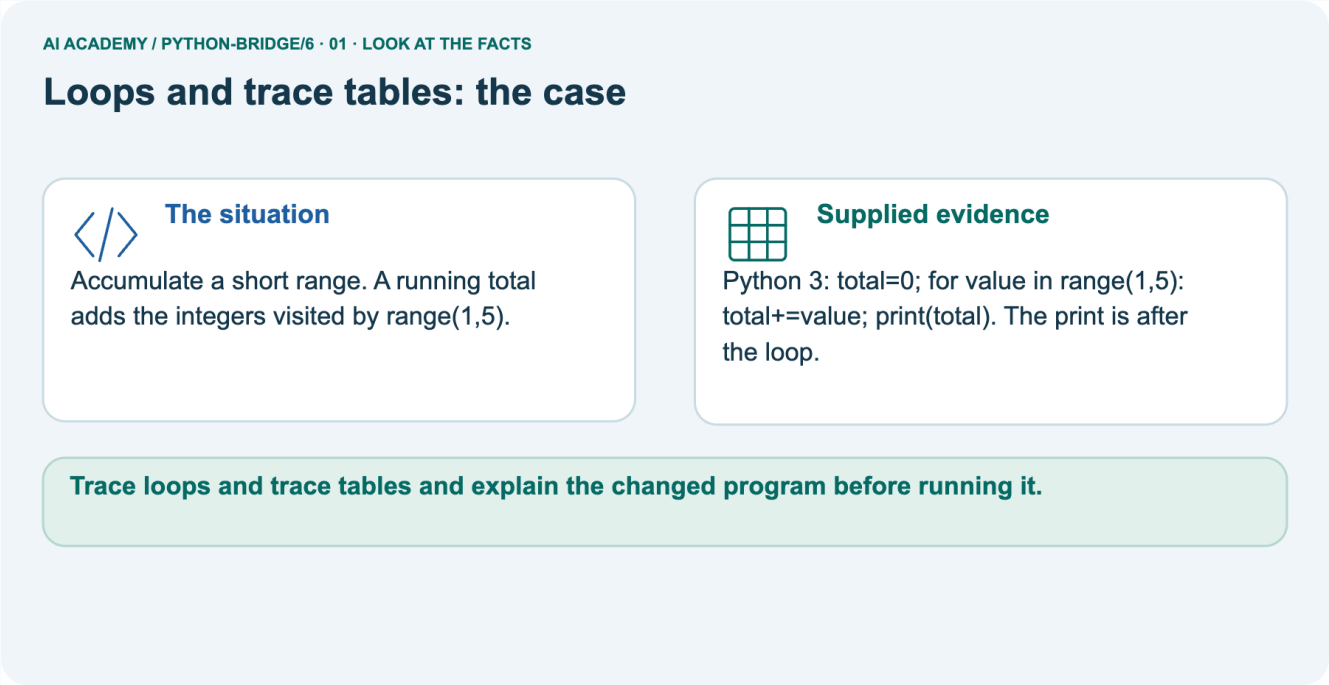
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Loops and trace tables graphic using its labels, deciding evidence and relevant limitation.

01. Loops and trace tables: the case

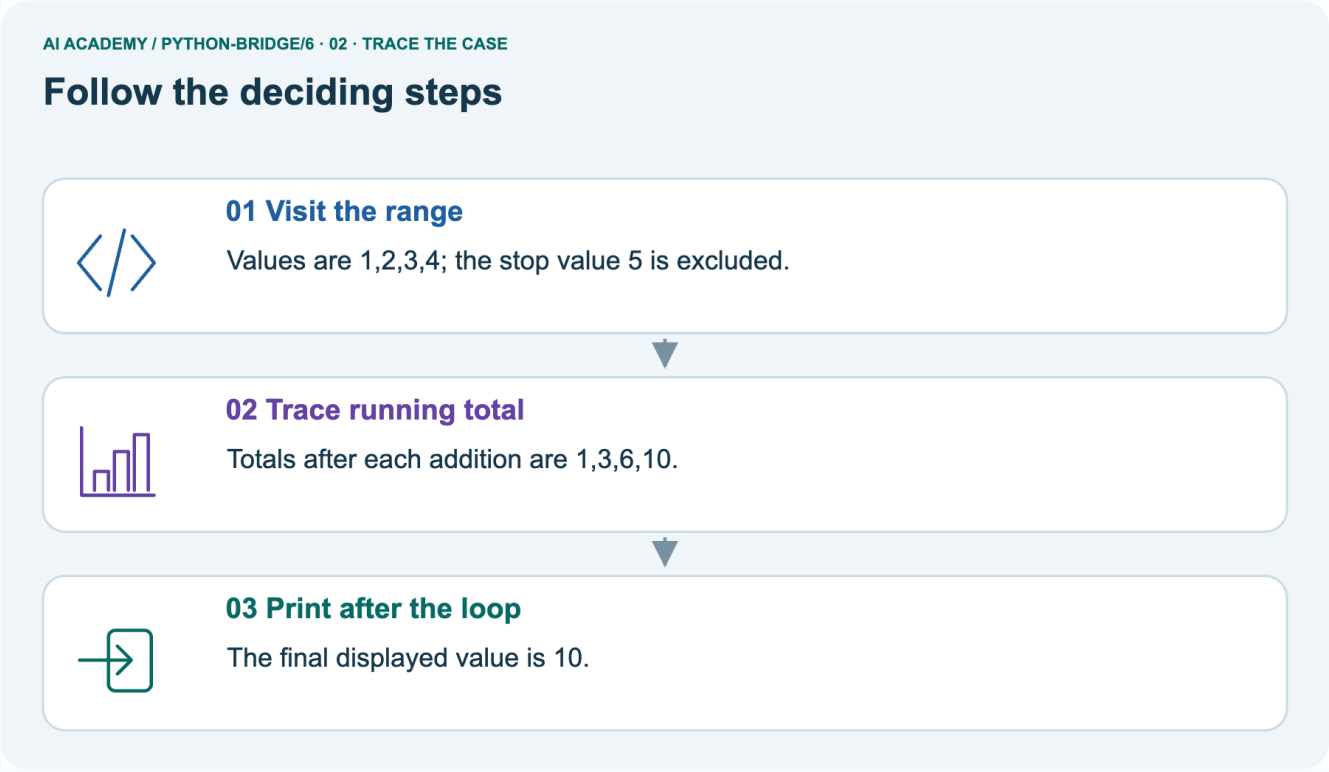


Follow the labelled picture with the matching explanation. Open the vector diagram

Accumulate a short range. A running total adds the integers visited by range(1,5). Supplied facts: Python 3: total=0; for value in range(1,5): total+=value; print(total). The print is after the **loop**.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

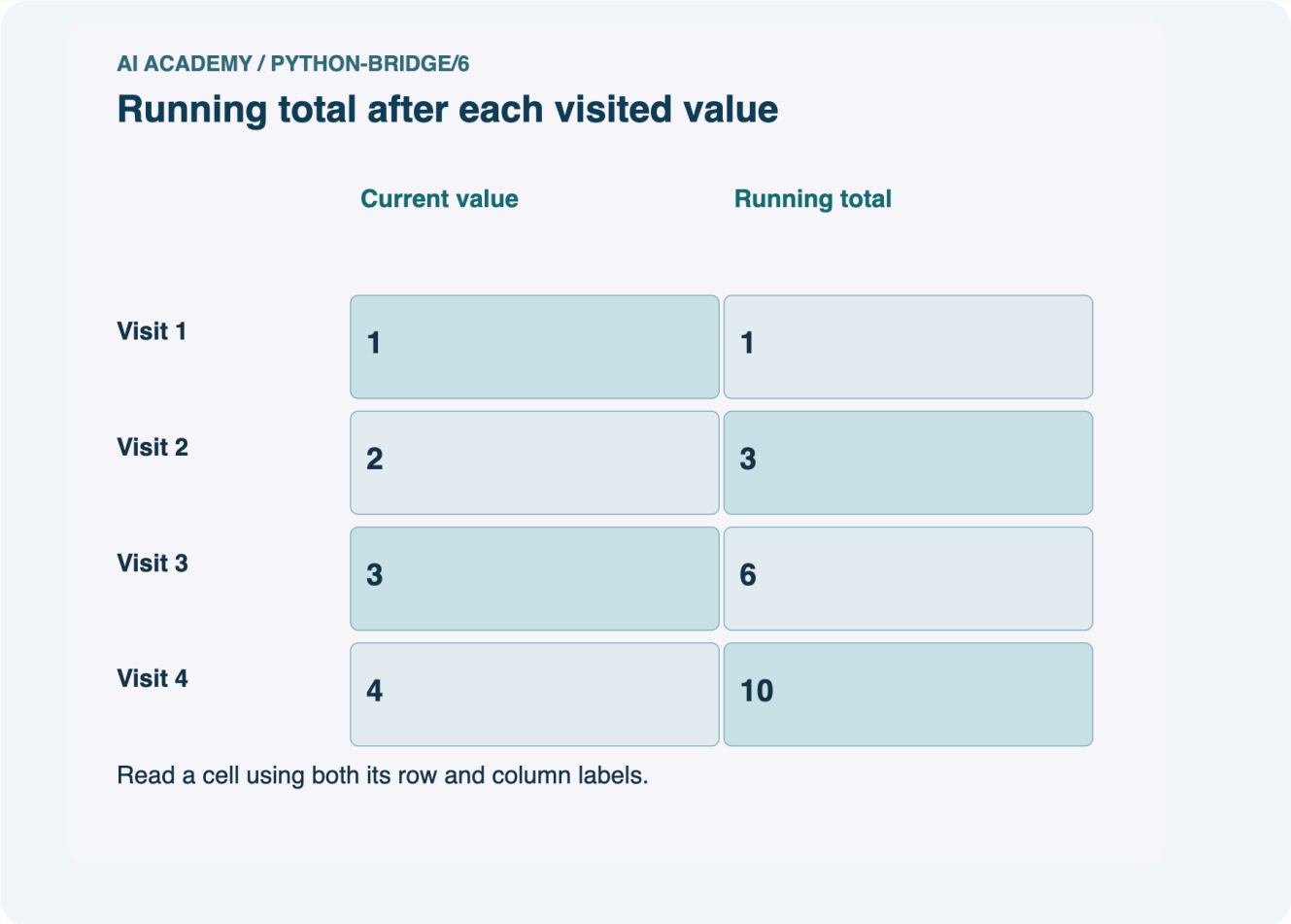


Follow the labelled picture with the matching explanation. Open the vector diagram

Visit the range: Values are 1,2,3,4; the stop value 5 is excluded. → Trace running total: Totals after each addition are 1,3,6,10. → Print after the **loop**: The final displayed value is 10.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Running total after each visited value



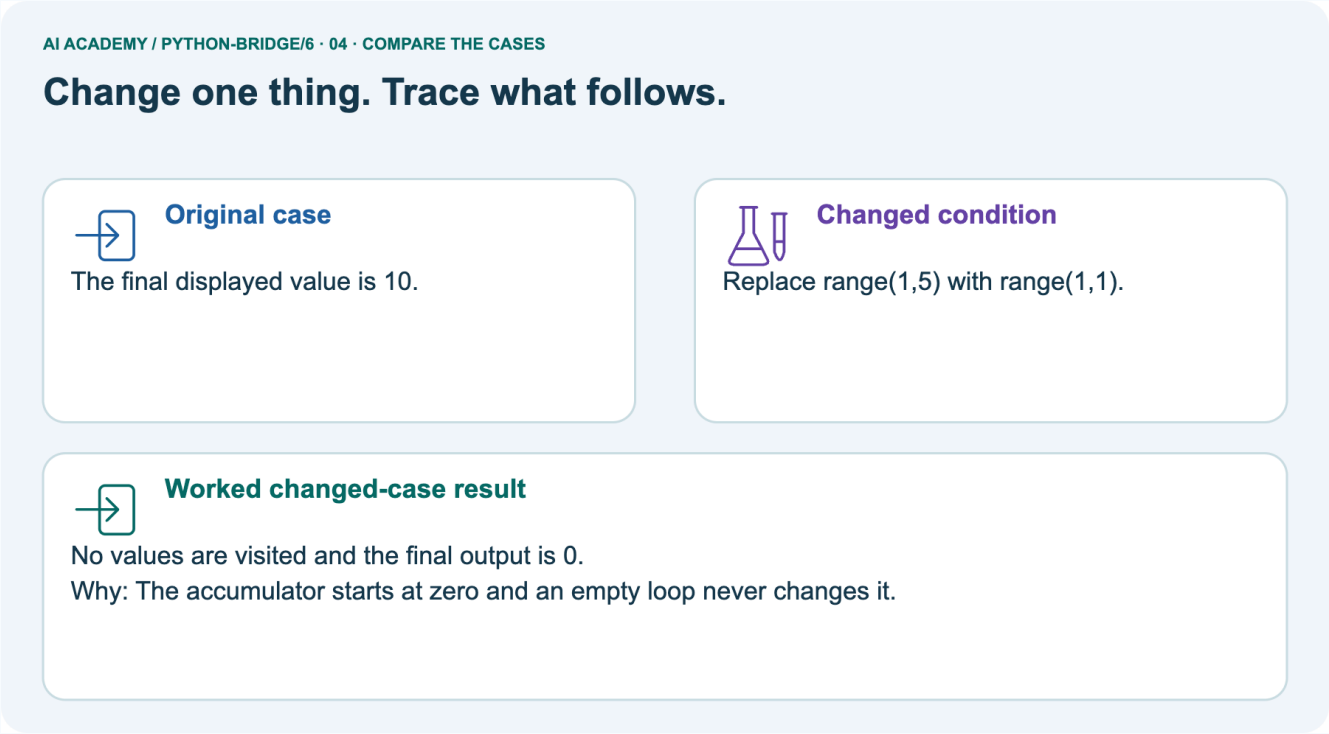
Follow the labelled picture with the matching explanation. Open the vector diagram

Running total after each visited value. Visit 1: Current value = 1, Running total = 1. Visit 2: Current value = 2, Running total = 3. Visit 3: Current value = 3, Running total = 6.

Visit 4: Current value = 4, Running total = 10.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



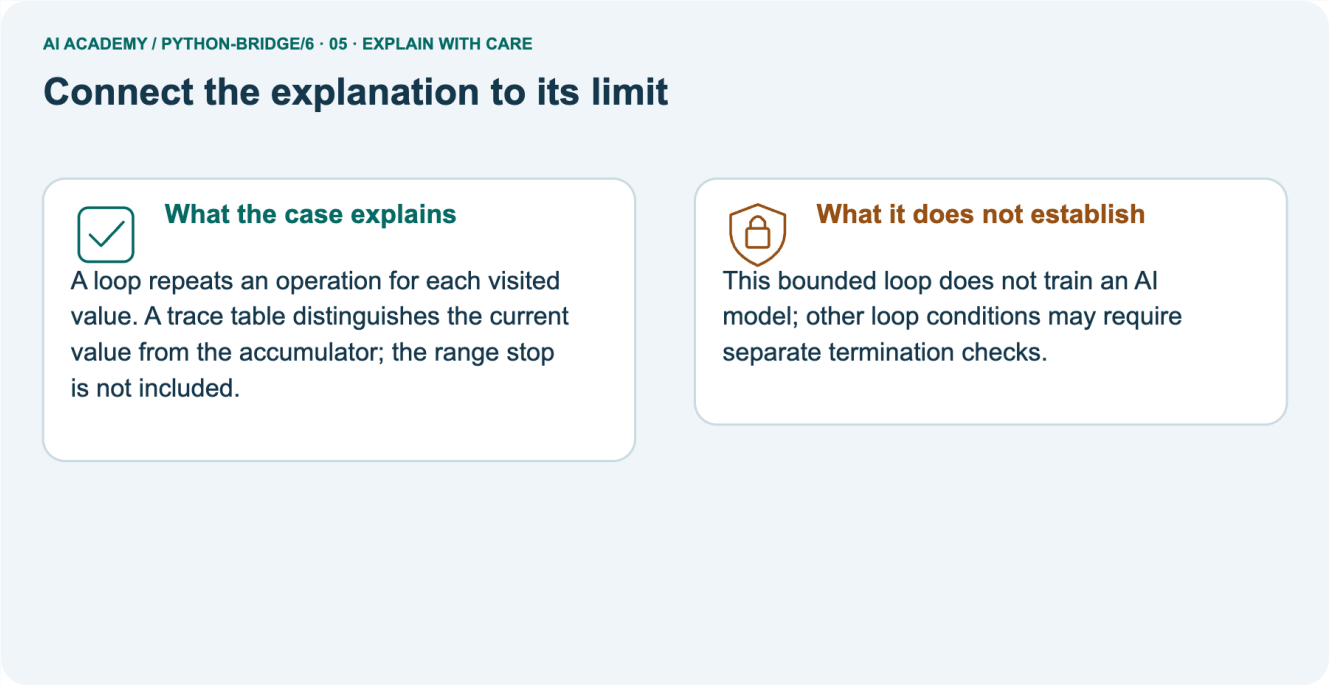
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The final displayed value is 10. Changed condition: Replace range(1,5) with range(1,1). Expected result: No values are visited and the final **output** is 0.

Reason: The **accumulator** starts at zero and an empty **loop** never changes it.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



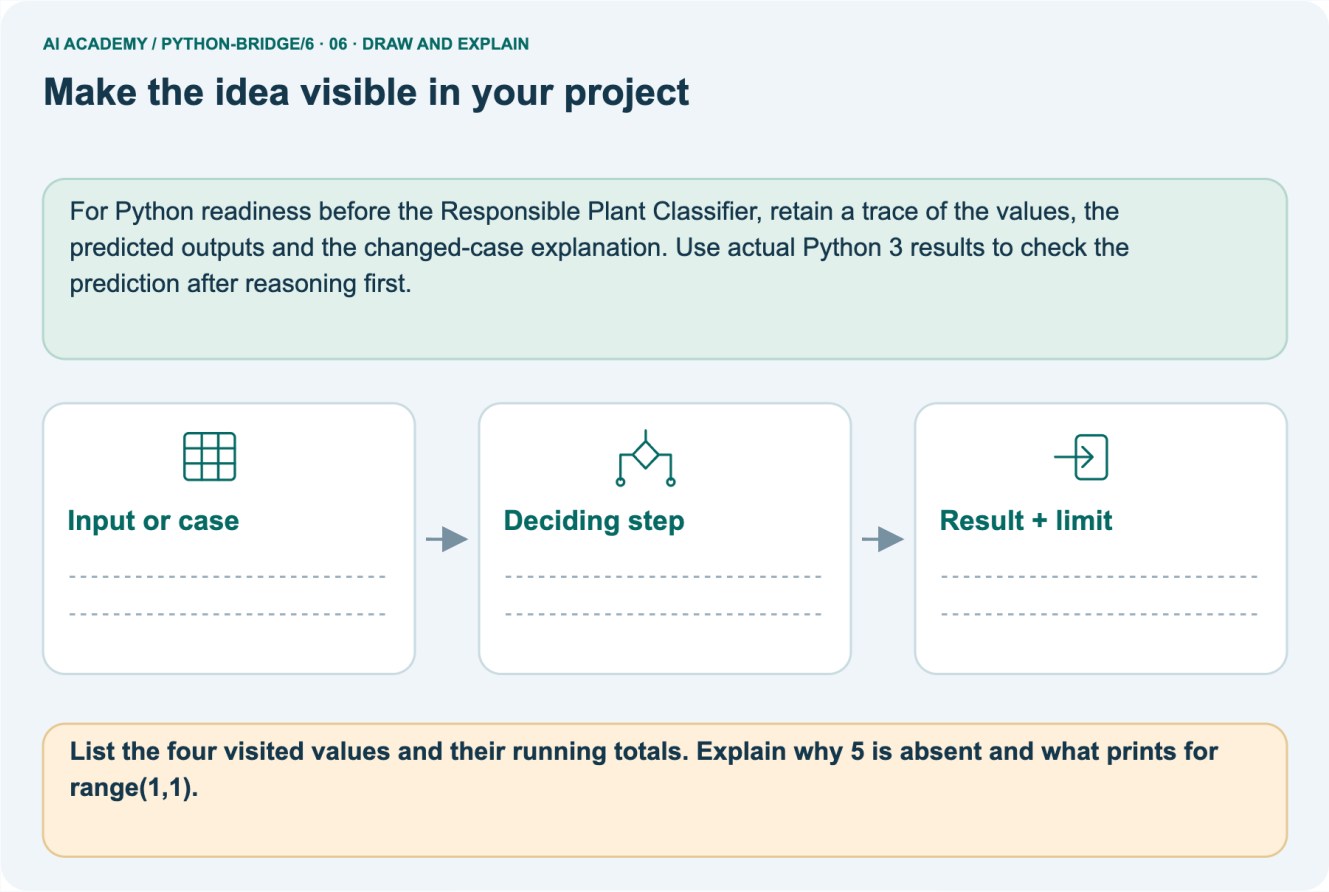
Follow the labelled picture with the matching explanation. Open the vector diagram

A **loop** repeats an operation for each visited value. A **trace table** distinguishes the current value from the **accumulator**; the range stop is not included.

Limit: This bounded **loop** does not train an AI model; other **loop** conditions may require separate termination checks.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: List the four visited values and their running totals. Explain why 5 is absent and what prints for range(1,1).

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.